"Field Theories of Condensed Matter Systems"
by E. Fradkin

CHAPTER 5

Sigma Models, Topological Terms and Spin Liquids

5.1 Generalized Spin Chains and Haldane's Conjecture

The phenomenology which emerges from the spin one-half Heisenberg antiferromagnetic chain is quite striking: no long range order, there are gapless states in particular, gapless spinless fermions (which, in the Heisenberg picture are solitons). From the point of view of the Hubbard model, the Heisenberg model occurs at infinite coupling where the charge-bearing degrees of freedom get a gap infinitely large. Thus spin and charge degrees of freedom are separated and the spin sector is at a critical point. This phenomenology inspired Anderson [Anderson 87] to propose a similar picture for the two-dimensional systems, the Resonating Valence Bond (RVB) picture. However, most of this picture surely should not generalize. Critical points are not generic and, in general, it is not possible to have gapless states without the spontaneous breaking of a continuous symmetry. Thus, the one-dimensional spin one-half case may be more the exception than the rule. For instance, it may be possible that the system is in a state without long-range order, which is likely to be massive. For this reason, it is important to consider generalizations of the Heisenberg model. This problem has been studied extensively. Two different approaches have been considered in one dimension (a) enlarging the representation (higher spin, same symmetry group SU(2)) and (b) higher symmetry groups (SU(N), for instance).

Haldane considered the generalization to higher spin but keeping the symmetry group to be SU(2) [Haldane 83] and [Haldane 85]. He first considered the *large spin limit* which should have semiclassical character. He showed that in this limit the effective Lagrangian was *almost* the Lagrangian of the quantum non-linear sigma model. That the non-linear sigma model should appear in a *semiclassical* ($S \to \infty$) limit should be of no surprise: one finds the same answer in mean-field theory. But there is something wrong with this picture.

The non-linear sigma model is known to have no long-range order and, in fact, it has a finite correlation length [Polyakov 75]. Thus if the sigma model truly was the infrared limit of the Heisenberg model, it could not possibly be a critical system, at least for S sufficiently large. Haldane found that this is indeed the case for spin systems in which S is an *integer*. For *half-integer* spins, he found that, in addition to the sigma model, there is an extra term which changes the physics drastically. The extra term turned out to be proportional to a topological invariant; the winding number or Pontryagin index of the (smooth) spin configuration. Thus it would appear that integer and half-integer *spin-chains* behave rather differently.

"Spin" systems with other symmetry groups have also been considered. These include SU(N) generalizations of the (SU(2)) Heisenberg model for various representations of the group. Affleck studied *a large-N limit* in which he was able to show that the ground state does not have long-range order and that there are no gapless states [Affleck 85]. However, other SU(N) generalizations of the Heisenberg model have been considered. For special choices of parameters, these systems are integrable (in the Bethe-Ansatz sense), and they are also at a critical point [Babudjian 1986]. Their critical behavior is however different from the one we discussed in the Heisenberg case. Thus, it appears that, at least in one dimension, these systems are either critical or in a *disordered state*, i.e. a state without long-range order and with only short-range spin correlations.

Let us first discuss the spin-s quantum Heisenberg chain. I will do so by introducing a path-integral method for spin systems which *does* generalize to higher dimensions, groups, representations, etc.

5.2 Path-Integrals for Spin Systems: The Single Spin Problem

In Chapter 3 we developed a path-integral method for Fermi systems of the Hubbard type (i.e. with local interactions). Using a Hubbard-Stratonovich transformation we were able to derive an effective action for the low-energy degrees of freedom, the spin fluctuations. The result was a path-integral representation of the long-range spin fluctuations, the quantum mechanical non-linear sigma model.

We also showed that, in the strong coupling limit, the half-filled Hubbard model maps onto the quantum Heisenberg model. In this limit the "band" fermions are bound into localized spins. There is no motion of fermions, the gap for charge fluctuations is infinitely large. It is natural to ask for an alternative derivation of the effective action for the spin fluctuations which should not be based on the weak coupling mean-field theory, as we did in Chapter 3. Also we will now be careful enough to keep terms of topological significance.

We begin with the discussion of an extremely simple system: a spin s degree of freedom coupled to an external field through a Zeeman term. From

the standard treatment in elementary quantum mechanics [Baym 74] we know that the $(2s+1)$-fold degeneracy is lifted by the Zeeman interaction, resulting in $(2s+1)$ non-degenerate levels. The path-integral will enable us to study the evolution operator between arbitrary initial and final states.

There are several published path-integral treatments of spin degrees of freedom. They all share the feature that they deal with coherent states rather than the more familiar complete states [Schulman 81]. The method of coherent states has been extensively reviewed by A. Perelonov [Perelomov 1986]. We will use a special version of the method of coherent states which keeps the spin symmetry intact, first introduced by Wiegmann [Wiegmann 88] and by Fradkin and Stone [Fradkin 88].

Let us begin by describing the Hilbert space. It is very simple. We have $(2s+1)$ states which transform like a spin-s representation of SU(2). Let $|0\rangle$ denote the *highest weight state* in this representation

$$|0\rangle = |S, S\rangle. \tag{5.2.1}$$

This state is an eigenstate of both S_3, the (only) diagonal generator of SU(2), and of the quadratic Casimir invariant $\vec{S}^2$

$$S_3|0\rangle = s|0\rangle, \tag{5.2.2}$$

$$\vec{S}^2|0\rangle = s(s+1)|0\rangle. \tag{5.2.3}$$

Consider now the state $|\vec{n}\rangle$ labeled by the unit vector $\vec{n}$ which is obtained by the rotation

$$|\vec{n}\rangle = e^{i\theta(\vec{n}_0\times\vec{n})\cdot\vec{S}}|S, S\rangle \tag{5.2.4}$$

where $\vec{n}_0$ is a unit vector along the quantization axis, θ is the co-latitude

$$\vec{n}\cdot\vec{n}_0 = \cos\theta \tag{5.2.5}$$

and S_i $(i = 1, 2, 3)$ are the (three) generators of SU(2) in the spin-s representation (see Fig. 5.1). For a review of SU(2) and its representations, see, for instance, Georgi [Georgi 82].

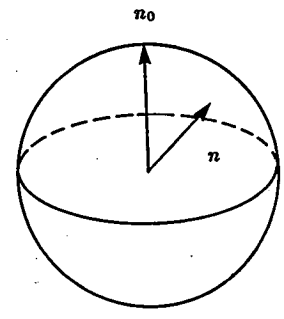


Figure 5.1 The unit sphere S_2 and the unit vectors $\vec{n}_0$ and $\vec{n}$.

The state $|\vec{n}\rangle$ can be expanded in a complete basis of the spin-s irreducible representation $\{|S,M\rangle\}$ where M labels the eigenvalue of S_3

$$S_3|S,M\rangle = m|S,M\rangle, \tag{5.2.6}$$

$$\vec{S}^2|S,M\rangle = s(s+1)|S,M\rangle \tag{5.2.7}$$

and $-s \leq m \leq s$. The coefficients of the expansion are the representation matrices $D^{(S)}(\vec{n})_{MS}$

$$|\vec{n}\rangle = \sum_{M=-S}^{S} D^{(S)}(\vec{n})_{MS}|S,M\rangle. \tag{5.2.8}$$

Clearly, there are many other rotations, differing from one another by multiplication on the right by rotations about the z-axis. This will give rise to the same state, except for an overall phase. In more formal terms, the observable states are in a one-to-one correspondence with the right co-sets SU(2)/U(1) where U(1), the phase, is generated by the *diagonal* generators of SU(2), that is S_3. In the language of differential geometry, the coherent states form a Hermitian line bundle associated with the Hopf, or monopole, principal bundle. The matrices $D^{(S)}$ do not form a group but rather satisfy

$$D^{(S)}(\vec{n}_1)D^{(S)}(\vec{n}_2) = D^{(S)}(\vec{n}_3)e^{i\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_3)S_3} \tag{5.2.9}$$

where $\vec{n}_1, \vec{n}_2$ and $\vec{n}_3$ are three arbitrary unit vectors and $\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_3)$ is the area of the spherical triangle with vertices at $\vec{n}_1$, $\vec{n}_2$ and $\vec{n}_3$ (see Fig. 5.2). Equation (5.2.9) is simply saying that the $D^{(S)}$-matrices form a group up to an element generated by the diagonal generators, the Cartan subalgebra. Since the sphere S_2 is a manifold without boundaries, the area of a spherical triangle is not uniquely defined. The shaded areas of the sphere in Fig. 5.2 (a)

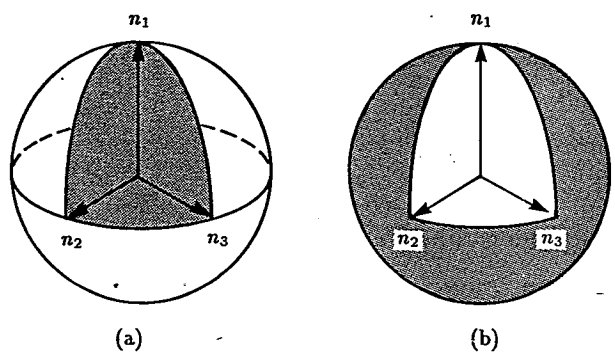


Figure 5.2 The spherical triangle with vertices at $\vec{n}_1$, $\vec{n}_2$ and $\vec{n}_3$. Its area is not unambiguously defined. "Inner" area (a) and "outer" area (b).

and (b) are equally good definitions of the area. The difference of the *oriented areas* is 4π. Since S_3 has eigenvalues equal to m, which is either an integer or a half-integer, this ambiguity has no physical manifestation since

$$e^{i4\pi m} = 1. \tag{5.2.10}$$

We can regard the requirement that the ambiguity in the definition of the area should lead to no physical consequences as the origin of the quantization of spin.

Other useful properties are the inner product $\langle\vec{n}_1|\vec{n}_2\rangle$

$$\begin{aligned}\langle\vec{n}_1|\vec{n}_2\rangle &= \langle 0|D^{(S)\dagger}(\vec{n}_1)D^{(S)}(\vec{n}_2)|0\rangle \\ &= e^{i\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_3)s}\left(\frac{1+\vec{n}_1\cdot\vec{n}_2}{2}\right)^s,\end{aligned} \tag{5.2.11}$$

the diagonal matrix elements of the generators $\vec{S}$

$$\langle\vec{n}|\vec{S}|\vec{n}\rangle = s\vec{n} \tag{5.2.12}$$

and the "resolution of unity" which is an expression of the identity operator $\hat{I}$ in terms of the coherent state operators $|\vec{n}\rangle\langle\vec{n}|$

$$\hat{I} = \int d\mu(\vec{n})|\vec{n}\rangle\langle\vec{n}|. \tag{5.2.13}$$

The integration measure $d\mu(\vec{n})$ is given by the invariant measure

$$d\mu(\vec{n}) = \left(\frac{2s+1}{4\pi}\right)d^3n\delta(\vec{n}^2-1). \tag{5.2.14}$$

We are now in a position to write down an expression for the path-integral in this coherent state representation. Its generalization to other groups is straightforward and has been given by Wiegmann. Let $H(\vec{S}) = \vec{B}\cdot\vec{S}$ be the Zeeman-like Hamiltonian for a spin system with one spin-s degree of freedom. I will consider the representation of the evolution operator in imaginary time

$$Z = \mathrm{Tr}\ e^{iHT} = \mathrm{Tr}\ e^{-\beta H}. \tag{5.2.15}$$

In other words, we are assuming that the initial and final states are identified. Let us split the imaginary time interval into N_t steps each of length δt and consider the limit $N_t \to \infty$ and $\delta t \to 0$ while keeping $N_t\delta t = \beta$ constant. As usual we make use of the Trotter formula

$$Z = \mathrm{Tr}\ e^{-\beta H} = \lim_{\substack{N_t\to\infty \\ \delta t\to 0}} \left(e^{-\delta t H}\right)^{N_t} \tag{5.2.16}$$

and insert the "resolution of identity", Eq. (5.2.13), at every intermediate time

$$Z = \lim_{\substack{N_t\to\infty \\ \delta t\to 0}} \left(\prod_{j=1}^{N_t}\int d\mu(\vec{n}_j)\right)\left(\prod_{j=1}^{N_t}\langle\vec{n}(t_j)|e^{-\delta t H}|\vec{n}(t_{j+1})\rangle\right) \tag{5.2.17}$$

with periodic boundary conditions. Here $\{t_j\}$ is a set of intermediate times in the imaginary-time interval $[0,\beta]$. Since δt is small we can approximate Eq.

(5.2.17)

$$Z = \lim_{\substack{N_t \to \infty \\ \delta t \to 0}} \left(\prod_{j=1}^{N_t} \int d\mu(\vec{n}_j) \right) \left(\prod_{j=1}^{N_t} [\langle \vec{n}(t_j) | \vec{n}(t_{j+1}) \rangle - \delta t \langle \vec{n}(t_j) | H | \vec{n}(t_{j+1}) \rangle] \right). \tag{5.2.18}$$

Within the same approximation we can write

$$\frac{\langle \vec{n}(t_j) | H | \vec{n}(t_{j+1}) \rangle}{\langle \vec{n}(t_j) | \vec{n}(t_{j+1}) \rangle} \approx \langle \vec{n}(t_j) | H | \vec{n}(t_j) \rangle + \mathcal{O}(\delta t). \tag{5.2.19}$$

Using the inner-product formula, Eq.(5.2.11), we get

$$\langle \vec{n}(t_j) | \vec{n}(t_{j+1}) \rangle = e^{i\Phi(\vec{n}(t_j), \vec{n}(t_{j+1}), \vec{n}_0)s} \left(\frac{1 + \vec{n}(t_j) \cdot \vec{n}(t_{j+1})}{2} \right)^s. \tag{5.2.20}$$

We now insert Eqs. (5.2.19) and (5.2.20) into Eq. (5.2.18) to find the (formal) path-integral

$$Z = \lim_{\substack{N_t \to \infty \\ \delta t \to 0}} \int \mathcal{D}\vec{n} e^{-\mathcal{S}_E[\vec{n}]} \tag{5.2.21}$$

where the measure $\mathcal{D}\vec{n}$ is given by

$$\mathcal{D}\vec{n} = \prod_{j=1}^{N_t} d\mu(\vec{n}(t_j)) \tag{5.2.22}$$

and the (exponentiated) Euclidean action is equal to

$$\begin{aligned} -\mathcal{S}_E[\vec{n}] = & is \sum_{j=1}^{N_t} \Phi(\vec{n}(t_j), n(t_{j+1}), \vec{n}_0) + s \sum_{j=1}^{N_t} \ln \left(\frac{1 + \vec{n}(t_j) \cdot \vec{n}(t_{j+1})}{2} \right) \\ & - \sum_{j=1}^{N_t} \langle \vec{n}(t_j) | H | \vec{n}(t_j) \rangle. \end{aligned} \tag{5.2.23}$$

In this derivation, we have assumed that the sequence of unit vectors $\{\vec{n}(t_j)\}$ are closed trajectories (because $\vec{n}(t_0) = \vec{n}(t_{N+1})$) on the sphere S_2 which are sufficiently smooth so that all the approximations of Eq. (5.2.19) make sense (see Fig. 5.3). This is not quite the case, as emphasized by Klauder [Klauder 79]. But these technicalities, as well as operator ordering problems, can be taken care of without affecting the physics. We will ignore these difficulties from now on. Our path integral will be as good a mathematical object as any other path integral.

The first term of the effective Euclidean action is complex. It leads to a sum over trajectories weighted by phases (even though we are working in imaginary time!) of the form

$$e^{is\mathcal{A}[\vec{n}]} \tag{5.2.24}$$

where $\mathcal{A}[\vec{n}]$ is the limit

$$\mathcal{A}[\vec{n}] = \lim_{\substack{N_t \to \infty \\ \delta t \to 0}} \sum_{j=1}^{N_t} \Phi(\vec{n}(t_j), \vec{n}(t_{j+1}), \vec{n}_0). \tag{5.2.25}$$

Since each term of this sum is the area of the spherical triangle with vertices at $\vec{n}(t_j)$, $\vec{n}(t_{j+1})$ and $\vec{n}_0$, the sum, i.e. the sum of these areas, is just equal to the total area of the cap Σ bounded by the trajectory Γ parametrized by $\vec{n}(t)$ (see Fig. 5.4). Once again, since S_2 has no boundaries, there are two caps Σ^+ and Σ^-. The oriented areas of Σ^+ and Σ^- also differ by 4π

$$\mathcal{A}(\Sigma^+) + \mathcal{A}(\Sigma^-) = 4\pi. \tag{5.2.26}$$

This is the same ambiguity we encountered before. It does not lead us to any observable effects since s is restricted to be an integer or a half-integer. The area of the cap Σ, say, Σ^+, is given by (in the limit $N_t \to \infty$, $\delta t \to 0$)

$$\mathcal{A}(\Sigma^+) = \int_0^1 d\tau \int_0^\beta dt \vec{n}(t,\tau) \cdot (\partial_t \vec{n}(t,\tau) \times \partial_\tau \vec{n}(t,\tau)) \equiv \mathcal{S}_{WZ}[\vec{n}] \tag{5.2.27}$$

where $\vec{n}(t,\tau)$ is an arbitrary, smooth parametrization of the cap Σ^+ bounded by Γ which satisfies the boundary conditions

$$\vec{n}(t,0) \equiv \vec{n}(t), \quad \vec{n}(t,1) \equiv \vec{n}_0, \quad \vec{n}(0,\tau) = \vec{n}(\beta,\tau), \tag{5.2.28}$$

where $t \in [0,\beta]$ and $\tau \in [0,1]$. Terms of this sort are generically called Wess-Zumino terms although sometimes they are also referred to as Chern-Simons terms.

We now proceed to take a naive continuum limit ($N_t \to \infty, \delta t \to 0$) and find from Eqs. (5.2.27) and (5.2.23) the Euclidean action

$$\mathcal{S}_E[\vec{n}] = -is\mathcal{S}_{WZ}[\vec{n}] + \frac{s\delta t}{4} \int_0^\beta dt (\partial_t \vec{n}(t))^2 + s \int_0^\beta dt \vec{B} \cdot \vec{n}(t) \tag{5.2.29}$$

where $\vec{B}$ is an external magnetic field.

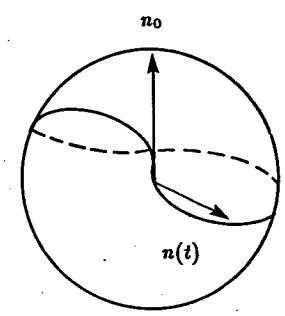


Figure 5.3 Closed smooth trajectories on S_2.

We can get back to real time x_0, with

$$t = ix_0 \quad \text{and} \quad \beta = iT \tag{5.2.30}$$

by writing

$$Z = \int \mathcal{D}\vec{n} e^{i\mathcal{S}_{\mathrm{M}}[\vec{n}]} \tag{5.2.31}$$

where $\mathcal{S}_{\mathrm{M}}[\vec{n}]$ is given by

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s\mathcal{S}_{\mathrm{wz}}[\vec{n}] + \left(\frac{s\delta t}{4}\right)\int_0^T dx_0(\partial_0\vec{n}(x_0))^2 - s\int_0^T dx_0\vec{B}\cdot\vec{n}(x_0). \tag{5.2.32}$$

This expression has a simple mechanical analogy. Let us imagine that $\vec{n}(x_0)$ is the position vector of a charged particle at time x_0. The particle has a small mass $m = \frac{s\delta t}{2}, (m \to 0)$ and is constrained to move on the surface of the unit sphere, S_2. A magnetic monopole with magnetic charge s is placed at the center of the sphere. The usual electromagnetic coupling gives a contribution to the action of the form [Landau 75].

$$\mathcal{S}_{\mathrm{em}} = \oint dx_0\vec{A}\cdot\frac{\partial\vec{n}}{\partial x_0} \tag{5.2.33}$$

where $\vec{A}$ is the vector potential at $\vec{n}$. In order to represent a monopole, the vector potential has to have a singular piece which describes the Dirac string. We can use Stoke's theorem to write $\mathcal{S}_{\mathrm{em}}$ in terms of a two-form instead of the one-form $\vec{A}$. Stoke's theorem simply says that $\mathcal{S}_{\mathrm{em}}$ is given by the flux of the magnetic monopole through the area of S_2 bounded by the trajectory Γ (see Fig. 5.5). This is nothing but the magnetic charge s of the monopole multiplied by the area of S_2 bounded by Γ, in other words, the cap Σ of Fig. 5.4. This is precisely identical to the first term in the action Eq. (5.2.32). Ideas of this sort were first popularized by Witten [Witten 83] in his discussion of Wess-Zumino terms (see also Stone [Stone 86]).

The magnetic monopole gives rise to a uniform radial magnetic field on the surface of the sphere with total flux equal to the magnetic charge s. It is well known that the eigenstates of such a particle are monopole spherical

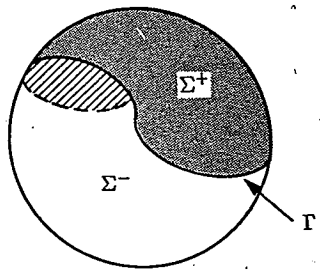


Figure 5.4 The trajectory Γ and the caps Σ^+ and Σ^-.

harmonics. The ground state is $(2s+1)$–fold degenerate and it is separated from the higher angular momentum states (i.e. Landau "orbits") by an energy gap which scales with the mass of the particle like $\frac{1}{m}$. Thus in the small mass limit $(m \to 0)$ the system is projected onto the ground state. In this way the subspace of the "lowest-Landau orbit" on a spherical geometry becomes identical to the space of the spin-s representation of SU(2). In retrospect, it would have been possible to describe spin in terms of the path-integral with Eq. (5.2.32) for its action directly, without reference to coherent states.

5.3 Path-Integral for Many-Spin Systems

It is trivial to generalize the one-spin problem to a many (or infinitely many!) spin situation. Once again, I will follow the treatment of Fradkin and Stone.

The Hilbert space of a many-spin system is just the tensor product of the Hilbert space of the individual spins. Let H be the Hamiltonian for a spin-s system on an *arbitrary lattice*

$$H = J\sum_{(\vec{r},\vec{r}')}\vec{S}(\vec{r})\cdot\vec{S}(\vec{r}') \tag{5.3.1}$$

where $(\vec{r},\vec{r}')$ are pairs of sites on that lattice. We can now use the identity $\langle\vec{n}|\vec{S}|\vec{n}\rangle = s\vec{n}$, to write down the imaginary-time action for the many-spin

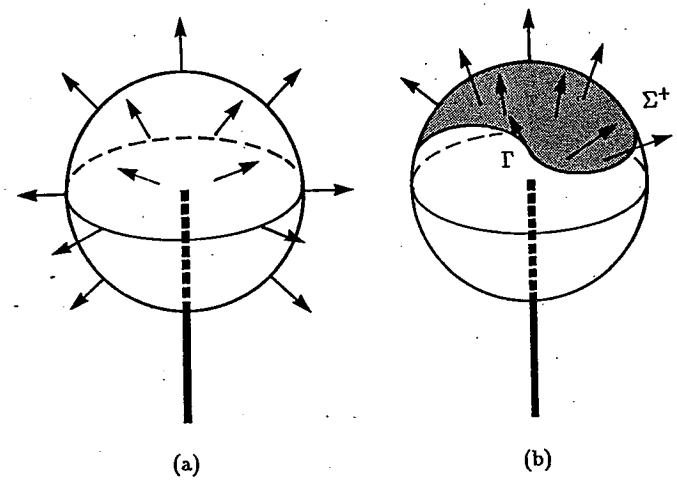


Figure 5.5 A magnetic monopole at the center of the unit sphere in (a) and the flux through the cap Σ^+ bounded by the trajectory Γ in (b). The thick line represents an infinitely long solenoid of infinitesimal thickness (Dirac string).

system

$$\mathcal{S}_{\mathrm{E}}[\vec{n}] = -is\sum_{\vec{r}}\mathcal{S}_{\mathrm{wz}}[\vec{n}(\vec{r}\,)] + \frac{m}{2}\int_0^\beta dt\sum_{\vec{r}}(\partial_t\vec{n}(\vec{r},t))^2 + \int_0^\beta dt\sum_{(\vec{r},\vec{r}')} Js^2\vec{n}(\vec{r},t)\cdot\vec{n}(\vec{r}\,',t) \tag{5.3.2}$$

where we are supposed to take the limit $m \to 0$ (it will be dropped from now on). The sums in Eq. (5.3.2) run over all the sites of the lattice. The first term is just the sum of the Wess-Zumino terms of the individual spins. Note that the only real time dependence enters through the Wess-Zumino terms.

We can Wick-rotate back to real time, $t = ix_0$, $\beta = iT$, and write the real-time action, $\mathcal{S}_{\mathrm{M}}$

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s\sum_{\vec{r}}\mathcal{S}_{\mathrm{wz}}[\vec{n}(\vec{r}\,)] - \int_0^T dx_0\sum_{(\vec{r},\vec{r}')} Js^2\vec{n}(\vec{r},x_0)\cdot\vec{n}(\vec{r}\,',x_0). \tag{5.3.3}$$

The effective action $\mathcal{S}_{\mathrm{M}}[\vec{n}]$ scale like s, the spin representation. Thus, in the large spin limit $s \to \infty$, the path-integral

$$Z = \int \mathcal{D}\vec{n}e^{i\mathcal{S}_{\mathrm{M}}[\vec{n}]} \tag{5.3.4}$$

should be dominated by the stationary points of the action $\mathcal{S}_{\mathrm{M}}[\vec{n}]$. This is the semiclassical limit. Corrections to the large-s limit can be arranged in an expression in powers of $\frac{1}{s}$. This is the content of the Holstein-Primakoff expansion [Holstein 40]. Note however that we did not make use of the semiclassical limit in order to derive the path-integral. Let us consider a number of cases of interest.

5.4 Quantum Ferromagnets

In this case we set $J = -|J|$. I will consider the case of an hypercubic lattice and restrict the sum over pairs of sites to nearest neighbors. The results can be generalized very easily to any other lattice.

I first make use of the constraint $\vec{n}^2 = 1$ to write the action in the form

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s\sum_{\vec{r}}\mathcal{S}_{\mathrm{wz}}[\vec{n}(\vec{r}\,)] - \frac{|J|s^2}{2}\sum_{<\vec{r},\vec{r}'>}\int_0^T dx_0[\vec{n}(\vec{r},x_0) - \vec{n}(\vec{r}\,',x_0)]^2 \tag{5.4.1}$$

up to an additive constant. Consider now the long-wavelength limit, in which $\vec{n}(\vec{r},x_0)$ is a smooth function of the spatial coordinates. If we denote by a_0 a short-distance cutoff (i.e. the lattice spacing) we can write the continuum action

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = \int d^dx\frac{s}{a_0^d}\mathcal{S}_{\mathrm{wz}}[\vec{n}] - \frac{|J|s^2}{2a_0^{d-2}}\int d^dx\int_0^T dx_0\left(\nabla_i\cdot\vec{n}(\vec{x},x_0)\right)^2. \tag{5.4.2}$$

It is important to stress that the effective continuum action for the *quantum ferromagnet* does not have the non-linear sigma model form. This is natural since the non-linear sigma model leads to Goldstone bosons with a *linear* dispersion law. This is the correct result for antiferromagnetic magnons but not for ferromagnetic magnons. It is well known that ferromagnetic magnons have a quadratic dispersion relation [Bloch 30].

To see how does all this come about we will derive the classical equations of motion. We take care of the *local constraint*

$$\vec{n}^2(\vec{x},x_0) = 1 \tag{5.4.3}$$

by introducing the Lagrangian multiplier field $\lambda(\vec{x},x_0)$ which enforces the constraint in the path-integral through an extra term in the action

$$\mathcal{S}_{\mathrm{extra}}[\vec{n},\lambda] = \int d^dx\int_0^T dx_0\frac{\lambda(\vec{x},x_0)}{2}\left(\vec{n}^2(\vec{x},x_0) - 1\right). \tag{5.4.4}$$

The classical equations of motion result from demanding that the total action

$$\mathcal{S}_{\mathrm{tot}}[\vec{n},\lambda] = \mathcal{S}_{\mathrm{M}}[\vec{n}] + \mathcal{S}_{\mathrm{extra}}[\vec{n},\lambda] \tag{5.4.5}$$

be stationary

$$\delta\mathcal{S}_{\mathrm{tot}} = 0. \tag{5.4.6}$$

The variation of the Wess-Zumino action is very simple. Indeed $\mathcal{S}_{\mathrm{wz}}$ is essentially the area of the sphere bounded by the trajectory $\vec{n}(\vec{x},x_0)$ (at each point $\vec{x}$) on the spin manifold (the two-sphere S_2). Thus the variation $\delta\mathcal{S}_{\mathrm{wz}}$ due to a small change in the trajectory $\delta\vec{n}$ is simply equal to

$$\delta\mathcal{S}_{\mathrm{wz}} = \delta\vec{n}\cdot(\vec{n}\times\partial_0\vec{n}). \tag{5.4.7}$$

Hence, we get the classical equations of motion

$$\frac{\delta\mathcal{S}_{\mathrm{tot}}}{\delta\vec{n}} = \nabla_i\left(\frac{\delta\mathcal{S}_{\mathrm{tot}}}{\delta\nabla_i\vec{n}}\right) \tag{5.4.8}$$

supplemented by the constraint Eq. (5.4.3). More explicitly, we get

$$\frac{s}{a_0^d}\vec{n}\times\partial_0\vec{n} + \lambda\vec{n} = -\frac{|J|s^2}{a_0^{d-2}}\nabla^2\vec{n}. \tag{5.4.9}$$

The classical value of the Lagrange field λ can be evaluated by computing the scalar product of Eq. (5.4.9) with $\vec{n}$. The result is

$$\lambda = -\frac{|J|s^2}{a_0^{d-2}}(\vec{n}\cdot\nabla^2\vec{n}). \tag{5.4.10}$$

Substituting Eq. (5.4.10) back in Eq. (5.4.9) I get the equation of motion for the quantum ferromagnet

$$\frac{s}{a_0^d}\vec{n}\times\partial_0\vec{n} + \frac{|J|s^2}{a_0^{d-2}}\left(\nabla^2 - (\vec{n}\cdot\nabla^2\vec{n})\right)\vec{n} = 0. \tag{5.4.11}$$

By using elementary algebra as well as Eq. (5.4.3), this equation can be brought to the form

$$\partial_0\vec{n} = |J|sa_0^2\vec{n}\times\nabla^2\vec{n}. \tag{5.4.12}$$

This equation is known as the Landau-Lifshitz equation. The derivation shown here is due to M. Stone.

The Landau-Lifshitz equation has several interesting properties. It is a non-linear equation with first-order time derivatives and second-order space derivatives. Thus the solutions of Eq. (5.4.12) have a quadratic dispersion law, as they should. The spins move in a precessional fashion with an angular velocity $\vec{\Omega}$ given by

$$\vec{\Omega} = -|J| s a_0^2 \nabla^2 \vec{n}. \tag{5.4.13}$$

The Landau-Lifshitz equations can be solved in the linear regime. Let us parametrize $\vec{n}$ by the components

$$\vec{n} = \begin{pmatrix} \vec{\pi} \\ \sigma \end{pmatrix} \tag{5.4.14}$$

where σ and π_i, $(i = 1, 2)$ satisfy the constraint

$$\sigma^2 + \vec{\pi}^2 = 1. \tag{5.4.15}$$

The (linearized) Landau-Lifshitz equations are

$$\begin{aligned} \partial_0 \pi_1 &\approx -|J| s a_0^2 \nabla^2 \pi_2, \\ \partial_0 \pi_2 &\approx +|J| s a_0^2 \nabla^2 \pi_1, \end{aligned} \tag{5.4.16}$$

to leading order in $\vec{\pi}$. From Eq. (5.4.16) we find the dispersion relation for ferromagnetic spin-waves

$$|p_0| \approx |J| s a_0^2 |\vec{p}|^2 \tag{5.4.17}$$

which is known as Bloch's law.

5.5 Effective Action for One-Dimensional Quantum Antiferromagnets

We will not consider here frustrated systems. Thus and for the sake of simplicity, we will consider the case of quantum antiferromagnets on *bipartite* lattices, such as the hypercubic lattice. We will see that, unlike the case of the ferromagnets, the effective low-energy action is *different* for one-dimensional systems and for higher dimensional cases such as the square and cubic lattices. In all cases we will find a non-linear sigma model, in agreement with our previous discussion (see Chapter 3) based on a mean-field weak coupling treatment of the Hubbard model. But we will get more. For the spin-chain case we will find that the action has an extra term, a topological term.

The starting point will be, once again, the real-time action of Eq. (5.3.3) with a nearest neighbor *antiferromagnetic* coupling constant $J > 0$. Since we expect that *at least* the short-range order should have Néel character, it is natural to consider the *staggered* and *uniform* components of the spin-field $\vec{n}$. This construction, as is, works only for two-sublattice systems close to a Néel state, although it is possible to generalize it to other cases.

Consider a spin chain with an *even* number of sites N occupied by spin–s degrees of freedom. The sites of the lattice are labeled by an integer $j = 1, \ldots, N$. The real time action is

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s \sum_{j=1}^{N} \mathcal{S}_{\mathrm{wz}}[\vec{n}(j)] - \int_0^T dx_0 \sum_{j=1}^{N} J s^2 \vec{n}(j, x_0) \cdot \vec{n}(j+1, x_0) \tag{5.5.1}$$

where we have assumed periodic boundary conditions. Since we expect to be close to a Néel state, we will stagger the configuration

$$\vec{n}(j) \to (-1)^j \vec{n}(j). \tag{5.5.2}$$

On a bipartite lattice, the substitution of Eq. (5.5.2) will change the sign of the *exchange term* of the action to a *ferromagnetic* one. The Wess-Zumino terms are *odd* under the replacement of Eq. (5.5.2) and thus get *staggered*. Thus, it is the Wess-Zumino term, a purely quantum mechanical effect, which will distinguish ferromagnets from antiferromagnets. After staggering the spins we get, up to an additive constant,

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s \sum_{j=1}^{N} (-1)^j \mathcal{S}_{\mathrm{wz}}[\vec{n}(j)] - \frac{J s^2}{2} \int_0^T dx_0 \sum_{j=1}^{N} (\vec{n}(j, x_0) - \vec{n}(j+1, x_0))^2 . \tag{5.5.3}$$

We now split the (staggered) spin field $\vec{n}$ into a slowly varying piece $\vec{m}(j)$, the order parameter field, and a small rapidly varying part, $\vec{l}(j)$, which roughly represents the average spin [Affleck 88]. Hence, we write

$$\vec{n}(j) = \vec{m}(j) + (-1)^j a_0 \vec{l}(j). \tag{5.5.4}$$

The constraint $\vec{n}^2 = 1$ and the requirement that the order-parameter field $\vec{m}$ should obey the same constraint, $\vec{m}^2 = 1$, demand that $\vec{m}$ and $\vec{l}$ be orthogonal vectors

$$\vec{m} \cdot \vec{l} = 0. \tag{5.5.5}$$

The Wess-Zumino term is rewritten

$$s \sum_{j=1}^{N} (-1)^j \mathcal{S}_{\mathrm{wz}}[\vec{n}(j)] = s \sum_{r=1}^{N/2} (\mathcal{S}_{\mathrm{wz}}[\vec{n}(2r)] - \mathcal{S}_{\mathrm{wz}}[\vec{n}(2r-1)]) \tag{5.5.6}$$

which, by making use of the approximation

$$\begin{aligned} \vec{n}(2r) - \vec{n}(2r-1) =& \vec{m}(2r) - \vec{m}(2r-1) + a_0(\vec{l}(2r) + \vec{l}(2r-1)) \\ =& a_0 \left(\partial_1 \vec{m}(2r) + 2\vec{l}(2r) \right) + \mathcal{O}(a_0^2), \end{aligned} \tag{5.5.7}$$

becomes

$$s\sum_{j=1}^{N}(-1)^j\mathcal{S}_{\rm wz}[\vec{n}(j)] \approx s\sum_{r=1}^{N/2}\int_0^T dx_0\,\delta\vec{n}(2r,x_0)\cdot(\vec{n}(2r,x_0)\times\partial_0\vec{n}(2r,x_0))$$
$$\approx s\sum_{r=1}^{N/2}\int_0^T dx_0\left(a_0\partial_1\vec{m}(2r,x_0)+2a_0\vec{l}(2r,x_0)\right)$$
$$\times(\vec{m}(2r,x_0)\times\partial_0\vec{m}(2r,x_0))\,. \tag{5.5.8}$$

Thus, in the continuum limit, one finds

$$\lim_{a_0\to 0} s\sum_{j=1}^{N}(-1)^j\mathcal{S}_{\rm wz}[\vec{n}(j)] \approx \frac{s}{2}\int d^2x\;\vec{m}\cdot(\partial_0\vec{m}\times\partial_1\vec{m})+s\int d^2x\;\vec{l}\cdot(\vec{m}\times\partial_0\vec{m})\,. \tag{5.5.9}$$

Similarly, the continuum limit of the potential energy terms can also be found to be given by

$$\lim_{a_0\to 0}\frac{Js^2}{2}\sum_{j=1}^{N}\int_0^T dx_0\,(\vec{n}(j,x_0)-\vec{n}(j+1,x_0))^2 \approx$$
$$\frac{a_0Js^2}{2}\int d^2x\left((\partial_1\vec{m})^2+4\vec{l}^2\right). \tag{5.5.10}$$

Collecting terms we find a Lagrangian density involving both the order-parameter field $\vec{m}$ and the local spin-density $\vec{l}$

$$\mathcal{L}_{\rm M}(\vec{m},\vec{l}\,) = -2a_0Js^2\vec{l}^2+s\vec{l}\cdot(\vec{m}\times\partial_0\vec{m})-\frac{a_0Js^2}{2}(\partial_1\vec{m})^2+\frac{s}{2}\vec{m}\cdot(\partial_0\vec{m}\times\partial_1\vec{m})\,. \tag{5.5.11}$$

The fluctuations in the average spin density $\vec{l}$ can be integrated out. The result is the Lagrangian density of the non-linear sigma model

$$\mathcal{L}_{\rm M}(\vec{m}) = \frac{1}{2g}\left(\frac{1}{v_s}(\partial_0\vec{m})^2-v_s(\partial_1\vec{m})^2\right)+\frac{\theta}{8\pi}\epsilon_{\mu\nu}\vec{m}\cdot(\partial_\mu\vec{m}\times\partial_\nu\vec{m}) \tag{5.5.12}$$

where g and v_s are the coupling constant and spin-wave velocity respectively:

$$g=\frac{2}{s}, \tag{5.5.13}$$
$$v_s = 2a_0Js. \tag{5.5.14}$$

The last term has topological significance. We have chosen the normalization so that θ is given by

$$\theta = 2\pi s. \tag{5.5.15}$$

The tensor $\epsilon_{\mu\nu}$ is the usual Levi-Civita antisymmetric tensor in two dimensions.

Thus, apart from an anisotropy determined by the spin-wave velocity v_s and for the topological term, we find that the effective action for the low-frequency, long-wavelength fluctuation about a state with *short-range* Néel order is given by the non-linear sigma model. We have reached the same results within the weak-coupling mean-field theory of the half-filled Hubbard model of Chapter 3. Indeed, using that approach, it is possible to get even the topological term [Wen 88].

5.6 The Role of Topology

In the past section we reached the conclusion that the low-energy excitations of a one-dimensional quantum antiferromagnet with short-range Néel order can be described by the path-integral of a non-linear sigma model with a topological term

$$Z=\int\mathcal{D}\vec{m}\,\delta(\vec{m}^2-1)e^{iS_{\rm eff}[\vec{m}]} \tag{5.6.1}$$

with the effective action obtained from Eq. (5.5.12). Before considering the role of local quantum fluctuations, which are of fundamental importance here, we look at the role of the last term in the action, the topological term $\mathcal{S}_t^M$

$$\mathcal{S}_t^M = \frac{\theta}{8\pi}\int d^2x\epsilon_{\mu\nu}\vec{m}\cdot(\partial_\mu\vec{m}\times\partial_\nu\vec{m})\,. \tag{5.6.2}$$

Let us consider first the Euclidean sector of the theory (i.e. we are back to imaginary time $x_2=ix_0$) with the Lagrangian density $\mathcal{L}_{\rm E}$

$$\mathcal{L}_{\rm E}=\frac{1}{2g}\left(v_s(\partial_1\vec{m})^2+\frac{1}{v_s}(\partial_2\vec{m})^2\right)+i\frac{\theta}{8\pi}\epsilon_{ij}\vec{m}\cdot(\partial_i\vec{m}\times\partial_j\vec{m})\,. \tag{5.6.3}$$

We now define the Pontryagin index (or Winding Number) Q of the Euclidean space spin configuration $\{\vec{m}(x)\}$ by the expression

$$Q=\frac{1}{8\pi}\int d^2x\epsilon_{ij}\vec{m}\cdot(\partial_i\vec{m}\times\partial_j\vec{m})\,. \tag{5.6.4}$$

We impose the boundary condition that the Euclidean action $\int d^2x\mathcal{L}_{\rm E}[\vec{m}]$ is finite. This is equivalent to the requirement that asymptotically $\vec{m}$ becomes a *constant* vector $\vec{m}_0$ at space-time infinity

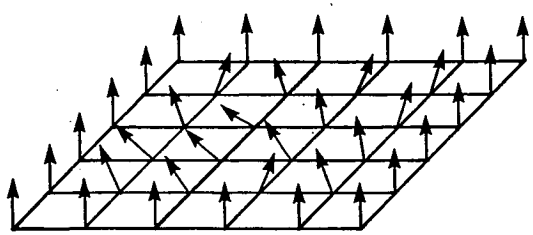
$$\lim_{|\vec{x}|\to\infty}\vec{m}(\vec{x})=\vec{m}_0. \tag{5.6.5}$$

Thus, topologically, Euclidean space-time is a sphere S_2 since the fields are identified with $\vec{m}_0$ at the point of infinity (Fig. 5.6). The order parameter manifold is also isomorphic to a sphere S_2, since the constraint $\vec{m}^2=1$ has to be satisfied everywhere. A field configuration $\vec{m}(x)$ with finite action is thus a smooth (differentiable) mapping from the S_2 of Euclidean space-time to the S_2 of the order parameter manifold (Fig. 5.7).

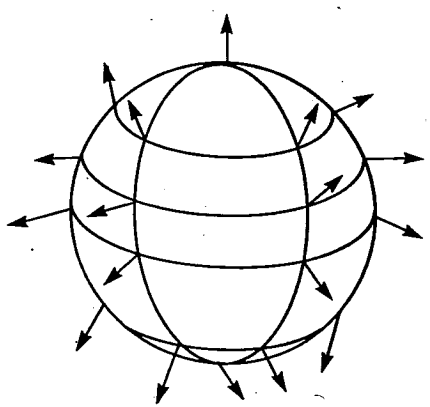
The Pontryagin Index $Q[\vec{m}]$ is the winding number in the sense that it counts how many times the spin configuration $\vec{m}$ has wrapped around the sphere, as can be checked by comparing the definition of Q, Eq. (5.6.4), with

the area formula of Eq. (5.2.27). We can make these ideas more concrete by considering a configuration $\vec{m}(x)$ representing an ***instanton*** (Fig. 5.8). Let the field at infinity point parallel to $\vec{m}_0$, the north pole of S_2. In the case of an instanton, the field near the origin points ***opposite*** to $\vec{m}_0$, i.e. in the direction of the south pole. Alternatively, we can look at the configuration on S_2. Here it looks like a magnetic monopole or a hairy ball (Fig. 5.9). The winding number Q of this configuration is determined by the area of the sphere divided by 4π (i.e. the "magnetic flux")

$$Q = \left(\frac{1}{4\pi}\right) 4\pi = +1. \tag{5.6.6}$$



(a)



(b)

Figure 5.6 A finite action spin configuration in Euclidean space-time (a) is isomorphic to one on the sphere S_2 (b).

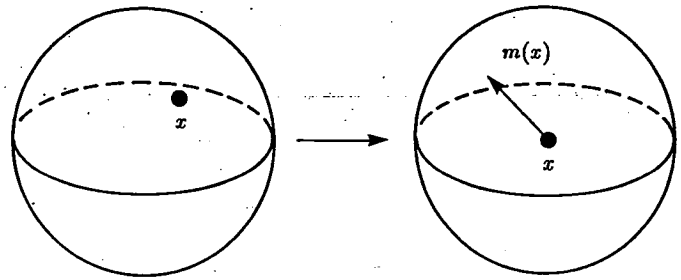


Figure 5.7 The mapping $\vec{m}(\vec{x})$.

Thus, an instanton has winding number $+1$. An anti-instanton has $Q = -1$. It is also possible to find multi-instanton configurations with arbitrary ***integral*** winding number Q.

We conclude that the smooth configurations $\vec{m}(x)$ can be classified according to their winding number. In other words, the configurations $\vec{m}(x)$ are mappings of S_2 into S_2 with ***homotopy*** classes classified by an integer, the Pontryagin Index Q. In mathematical terms the statement is

$$\Pi_2(S_2) = \mathcal{Z}. \tag{5.6.7}$$

Back in real time we can consider ***soliton*** configurations, such as the half-twist of Fig. 5.10. As time goes by, each spin traces a closed path on the sphere S_2 and hence it sweeps an area bound by the path. If we define that the area swept by a spin at $-\infty$ is equal to zero, we see that as we move from left to right the spins sweep an increasingly large area. At $+\infty$ the area swept is that of a full sphere, 4π. It is easy to see that Q is also equal to one for

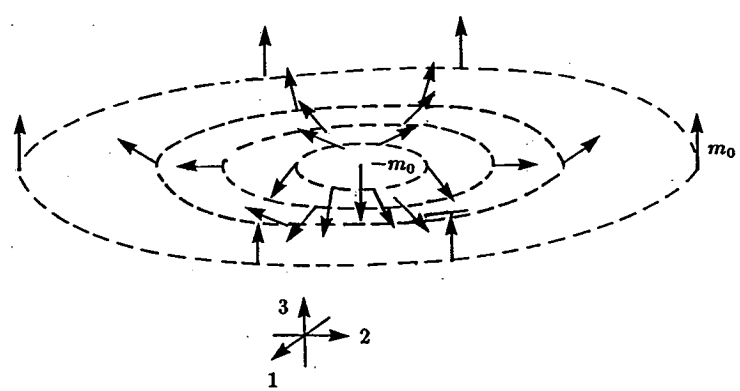


Figure 5.8 An instanton configuration in Euclidean space-time.

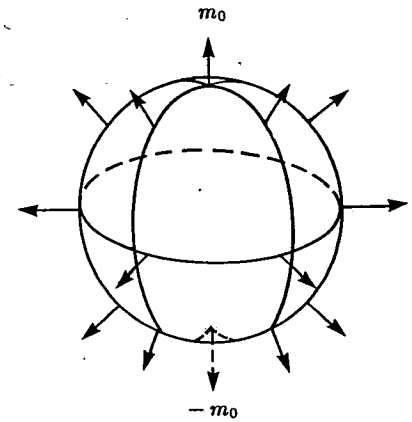


Figure 5.9 An instanton on S_2 has the same topology as a monopole.

the half-twist. At each point in space, the spins are coherently precessing and keeping their relative angles constant. In other words, the spins trace lines of longitude on a sphere. The global configuration still looks like a monopole and hence also has winding number +1 (Fig. 5.11).

The final conclusion is that the topological term, Eq. (5.6.2), is proportional to an integer Q. The action in the path-integral of Eq. (5.6.1) has a contribution equal to $(2\pi s)Q$ which should be added to the sigma-model term. Since s is an integer or a half-integer, we find that the extra, topological, term gives conbribution

$$e^{i2\pi sQ} = (-1)^{2sQ}. \tag{5.6.8}$$

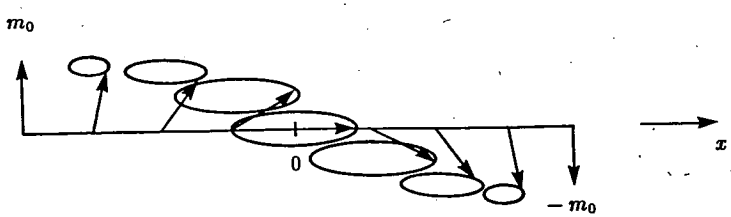


Figure 5.10 Half-twist soliton. The circles represent the precession of the spins.

Thus, if s is an integer, the spin-chain is described at low energies by the non-linear sigma model. For a half-integral s, each topological *class* contributes with a sign which is positive (negative) if the winding number Q is even (odd). Note that the sign does not depend on the actual value of the spin s, but only on whether it is an integer or half-integer. This means that the physics of this problem is *not* analytic in s: the integer and half-integer spin-chains fall in different universality classes. We will now see that this property implies a very important result, known as Haldane's conjecture, which states that the integer spin chains are massive (i.e. have a gap) while the half-integer chains are massless as in the spin one-half case.

5.7 Quantum Fluctuations: Renormalization Group

In the previous section we saw that the configuration space of the non-linear sigma model can be partitioned into classes classified by their winding numbers

$$Q = \frac{1}{8\pi}\int d^2x \epsilon_{abc}\epsilon_{ij} m_a \partial_i m_b \partial_j m_c \tag{5.7.1}$$

which are topological invariants. Thus, the partition function is a sum over distinct topological sectors

$$Z = \int \mathcal{D}\vec{m} e^{-S_E[\vec{m}]} = \sum_{Q=0}^{\infty} \int_Q \mathcal{D}\vec{m} e^{-S_0^E[\vec{m}]} e^{i2\pi sQ} \tag{5.7.2}$$

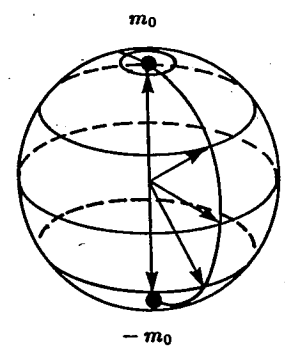


Figure 5.11 History of a half-twist.

where the subindex Q indicates that the path-integral is to be performed over configurations with a fixed winding number Q and $\mathcal{S}_0^E[\vec{m}]$ is the standard action of the non-linear sigma model

$$\mathcal{S}_0^E[\vec{m}] = \int d^2x \frac{1}{2g} (\nabla_i \vec{m})^2 \tag{5.7.3}$$

where space and time have been rescaled so as to have $v_s = 1$.

In this section we will consider the role of quantum fluctuations. We can do so by considering each topological class separately since these quantum fluctuations are local and do not alter the winding number. In other words, the winding number of a class of configurations cannot be changed by local fluctuations, since the former is a global property whereas the latter are purely local. Naturally, for this picture to hold it is necessary that the short distance (ultraviolet) and the long distance (infrared) of the theory remain separate. We will see that this is not the case in one-space-dimension. The behavior of the non-linear sigma model is dominated by *infrared* fluctuations. Thus the actual role, in detail, of topological sectors is unclear.

We will pretend that the fluctuations are local and reasonably small. This assumption amounts to a semiclassical treatment of the path-integral. Formally, this can be achieved only if the coupling constant g is small, i.e. in the limit $s \to \infty$. The standard perturbative treatment of the non-linear sigma model is thus equivalent, at low energies, to the $\frac{1}{s}$ expansion of the Heisenberg antiferromagnet [Haldane 83]. The classical action of the non-linear sigma model, Eq. (5.7.3), has a very important property: it is *scale invariant*. In other words the scale transformation

$$\begin{aligned} (x,t) &\to \lambda(x,t), \\ \vec{m} &\to \vec{m}, \end{aligned} \tag{5.7.4}$$

leaves the *action* invariant. Recall that $\vec{m}$ is dimensionless. The coupling constant g is also *dimensionless* in $(1+1)$ dimension. In higher dimensions, g is *dimensionful*. Let us define the *dimensionless coupling constant* u

$$u = g a_0^{2-d} \tag{5.7.5}$$

where d is the dimension of space-time. Thus the action now reads

$$\mathcal{S}_0^E[\vec{m}] = \frac{1}{2ua_0^{d-2}} \int d^dx \, (\nabla_i \vec{m})^2 \tag{5.7.6}$$

where $i = 1, \ldots, d$. For the sake of simplicity the discussion will be carried out in Euclidean space (i.e. imaginary time).

In Renormalization Group Theory [Wilson 74] the fact that the classical action is scale invariant means that $g = 0$ is a fixed point of the renormalization group (RG). I will define *a renormalization group transformation* by progressively integrating out the faster modes and obtaining an effective theory for the slower modes. In general, the sigma-field $\vec{m}$ will have Fourier components with momenta $\vec{p}$ ranging from the infrared ($|\vec{p}| \approx 0$) to the ultraviolet ($|\vec{p}| \approx \frac{1}{a_0}$). We can also use the constraint $\vec{m}^2 = 1$ to demand that one of the

components of the field $\vec{m}$, say m_3, has only fast components and be small [Kogut 79] . Let m_1 and m_2 be parametrized by m_3 and $\theta, (0 \le \theta \le 2\pi)$

$$m_1 = \sqrt{1 - m_3^2} \cos\theta, \quad m_2 = \sqrt{1 - m_3^2} \sin\theta \tag{5.7.7}$$

so as to solve the constraint $\vec{m}^2 = 1$. The Euclidean Lagrangian density now reads

$$\begin{aligned} \mathcal{L}_0^E &= \frac{1}{2ua_0^{d-2}} (\nabla_i \vec{m})^2 \\ &= \frac{1}{2ua_0^{d-2}} \left[(\nabla_i m_3)^2 + (1 - m_3^2)(\nabla_i \theta)^2 + \frac{(m_3 \nabla_i m_3)^2}{1 - m_3^2} \right]. \end{aligned} \tag{5.7.8}$$

Let us rescale m_3

$$m_3 = \sqrt{ua_0^{d-2}} \varphi \tag{5.7.9}$$

and write

$$\mathcal{L}_0^E = \frac{1}{2}(\nabla_i \varphi)^2 + \frac{1}{2ua_0^{d-2}}(1 - ua_0^{d-2}\varphi^2)(\nabla_i \theta)^2 + \frac{1}{2}\left(\frac{ua_0^{d-2}}{1 - ua_0^{d-2}}\right)(\varphi \nabla_i \varphi)^2. \tag{5.7.10}$$

We will be interested in the behavior for small g (i.e. small u). In this limit we can approximate $\mathcal{L}_0^E$ by the expression

$$\begin{aligned} \mathcal{L}_0^E =& \frac{1}{2}(\nabla_i \varphi)^2 + \frac{1}{2ua_0^{d-2}}(\nabla_i \theta)^2 - \frac{1}{2}\varphi^2 (\nabla_i \theta)^2 + \\ &+ \frac{1}{2} ua_0^{d-2} (\varphi \nabla_i \varphi)^2 + \frac{1}{2} u^2 a_0^{2(d-2)} \varphi^2 (\varphi \nabla_i \varphi)^2 + 0(u^3). \end{aligned} \tag{5.7.11}$$

Both φ and θ have Fourier components all the way from zero momentum up to the cutoff $\Lambda \sim \frac{1}{a_0}$ The behavior at large momenta $|\vec{p}| \sim \Lambda$ should not affect very strongly phenomena taking place for small values of $\vec{p}$. It is then natural to integrate out such fluctuations.

Consider the momentum shell $b\Lambda < |\vec{p}| < \Lambda$ with $b < 1$ and the fluctuations with momenta inside that shell (fast modes). We now will carry out the functional integral

$$\begin{aligned} &\int_{b\Lambda < |\vec{p}| < \Lambda} \mathcal{D}\varphi e^{-\mathcal{S}_0^E[\varphi,\theta]} = \\ &\int_{b\Lambda < |\vec{p}| < \Lambda} \mathcal{D}\varphi(\vec{p}) e^{-\frac{1}{2}\int d^dx [(\nabla_i \varphi)^2 + \frac{1}{ua_0^{d-2}}(\nabla_i \theta)^2 - \varphi^2 (\nabla_i \theta)^2 + \mathcal{O}(u)]} \end{aligned} \tag{5.7.12}$$

I will assume that θ is slowly varying and, hence, that its gradient, $(\nabla_i \theta)^2$, is small and it does not have Fourier components in the shell $b\Lambda < |\vec{p}| < \Lambda$

provided that $b \to 1$. Thus

$$\int_{b\Lambda<|\vec{p}|<\Lambda} \mathcal{D}\varphi(\vec{p}) e^{-\frac{1}{2}\int \frac{d^dp}{(2\pi)^d}\vec{p}^2|\varphi(\vec{p})|^2+\frac{1}{2}\int \frac{d^dp}{(2\pi)^d}|\varphi(\vec{p})|^2(\nabla_i\theta)^2} \approx \prod_{b\Lambda<|\vec{p}|<\Lambda} \left[\frac{2\pi}{\vec{p}^2-(\nabla\theta)^2}\right]^{\frac{1}{2}}. \tag{5.7.13}$$

The right-hand side of Eq. (5.7.13) can be exponentiated and approximated by the expression

$$\exp\left[\frac{1}{2}\int_{b\Lambda<|\vec{p}|<\Lambda} \frac{d^dp}{(2\pi)^d}\ln\frac{2\pi}{\vec{p}^2}+\frac{1}{2}(\nabla_i\theta)^2\int_{b\Lambda<|\vec{p}|<\Lambda}\frac{d^dp}{(2\pi)^d}\frac{1}{\vec{p}^2}\right]. \tag{5.7.14}$$

To lowest order in u, the main effects of integrating out the fast modes in φ are twofold: (a) a shift of energy and (b) a shift, or renormalization, of the coupling constant u. Indeed, we can recast Eqs. (5.7.12) and (5.7.13) into the effective Lagrangian density

$$\begin{aligned}\mathcal{L}^E_{\text{eff}} = &-\frac{1}{2}\int_{b\Lambda<|\vec{p}|<\Lambda}\frac{d^dp}{(2\pi)^d}\ln\frac{2\pi}{\vec{p}^2}+\\ &+\frac{1}{2}(\nabla_i\varphi)^2+\frac{1}{2}\left(\frac{1}{ua_0^{d-2}}-\int_{b\Lambda<|\vec{p}|<\Lambda}\frac{d^dp}{(2\pi)^d}\frac{1}{\vec{p}^2}\right)(\nabla_i\theta)^2-\frac{1}{2}\varphi^2(\nabla_i\theta)^2+\\ &+\mathcal{O}(u),\end{aligned} \tag{5.7.15}$$

with a momentum cutoff Λ' which has been *reduced* by b. Equivalently, the spatial cutoff a_0' has been *increased* by $\frac{1}{b}$:

$$\Lambda' = b\Lambda, \quad a_0' = \frac{a_0}{b}. \tag{5.7.16}$$

The effective Lagrangian density for the slower modes $\mathcal{L}^E_{\text{eff}}$ has the same form as the old Lagrangian density except for a constant shift, a new cutoff a_0' ($a_0' > a_0$) and a new renormalized coupling constant u' defined by

$$\frac{1}{u'a_0'^{d-2}} = \frac{1}{ua_0^{d-2}} - \int_{b\Lambda<|\vec{p}|<\Lambda}\frac{d^dp}{(2\pi)^d}\frac{1}{\vec{p}^2}. \tag{5.7.17}$$

After evaluating the integral, we get

$$\frac{1}{u'a_0'^{d-2}} = \frac{1}{ua_0^{d-2}} - \frac{S_d}{(2\pi)^d}\frac{1}{d-2}(1-b^{d-2})\Lambda^{d-2} \tag{5.7.18}$$

where S_d is the area of the d-dimensional unit sphere. Since $b \to 1$ and $a_0' = \frac{a_0}{b}$ we can write

$$-\ln b = \frac{da_0}{a_0} \tag{5.7.19}$$

and find the β-function

$$\beta(u) = a_0\frac{du}{da_0} \tag{5.7.20}$$

to be given by

$$\beta(u) = -\epsilon u + \frac{u^2}{2\pi} + \mathcal{O}(u^3) \tag{5.7.21}$$

for $\epsilon = d-2$ *small*.

In particular in $1+1$ dimensions ($d=2$) we find a *positive* β-function [Polyakov 75].

$$\beta(u) = \frac{u^2}{2\pi}. \tag{5.7.22}$$

This result means that as the cutoff a_0 is increased, and we look at longer and longer distances, the fluctuations *increase* the effective value of the coupling constant at such scales (see Fig. (5.12)). Thus, even though the *bare* coupling constant $u_0 \propto \frac{1}{s}$ may be initially small, as we consider the effective theory at lower energies we find that the effective coupling ("effective s") increases (decreases). From classical statistical mechanics we know that the sigma model at strong coupling (i.e. the classical Heisenberg ferromagnet at high temperatures) is disordered and has a finite correlation length. Thus, in the language of the quantum spin chains, we get that as the "effective s" decreases the semi-classical behavior gets wiped-out. Instead we find a state *without* spontaneous symmetry breaking and with short range correlations.

5.8 Asymptotic Freedom and Haldane's Conjecture

In the last section we found the result that the effective coupling constant of the non-linear sigma model in $(1+1)$ dimensions increases with the length scale. We have chosen to present this result in the form of a β–function, Eq. (5.7.22), which measures the change of the coupling constant u as the cutoff a_0 (the lattice constant) is increased and the fast degrees of freedom of the system are progressively integrated out. Alternatively, we could have kept the cutoff fixed and varied a physical scale such as the length L of the chain or an energy scale such as the temperature T.

At finite temperature T, the system can be viewed (in imaginary time) as a non-linear sigma model on a strip of length L (the linear size of the chain) and width $\frac{1}{T}$ with periodic boundary conditions in imaginary time. This is the standard statement that the partition function of a quantum system, with a

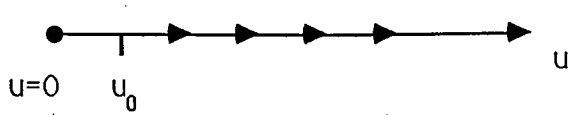


Figure 5.12 Renormalization Group Infrared Flow for one-dimensional Quantum Spin-Chains; u_0 is the bare coupling constant ($u_0 = \frac{2}{s}a_0^{2-d}$).

global symmetry, in d space dimensions is equivalent to a classical mechanics problem in $(d+1)$ dimensions with imaginary time being the extra dimension [Fradkin 78] (Fig. 5.13). The renormalization group of the last section can easily be generalized to an anisotropic system with spin wave velocity $v_s \neq 1$ which is kept fixed in the RG process.

We begin our RG process with some lattice constant a_0, bare coupling $u_0 \propto \frac{1}{s}$ and spin wave velocity v_s. As we integrate out degrees of freedom the effective coupling grows and the spatial cutoff increases. At some point, the cutoff a becomes of the order of $\frac{v_s}{T}$. At this point the quantum fluctuations are negligible since the cutoff is as large as the width of the strip and we have effectively a *non-linear* sigma model at finite temperature T. In turn the non-linear sigma-model, in imaginary time, is identical to the *classical* Heisenberg model in d space-time Euclidean dimensions. It can be easily proven that a classical Heisenberg model (or non-linear sigma model) in one dimension, like all one-dimensional classical systems with short-range interactions, has a finite correlation length ξ_c at all temperatures [Landau 75].

We can now ask how much does the effective coupling u differ from the bare coupling u_0 if the cutoff is changed from a_0 to $\bar{a}_0 \sim \frac{v_s}{T}$. The β–function tells us what is the dependence of u on the cutoff, at least for small enough u. The result of integrating the differential equation

$$\beta(u) \equiv a_0 \frac{du}{da_0} = \frac{u^2}{2\pi} \tag{5.8.1}$$

is

$$\frac{1}{u(\bar{a}_0)} = \frac{1}{u(a_0)} + \frac{1}{2\pi} \ln\left(\frac{a_0}{\bar{a}_0}\right). \tag{5.8.2}$$

By choosing $\bar{a}_0$ to be of the order of $\frac{1}{T}$

$$\bar{a}_0 = \frac{v_s}{T} \tag{5.8.3}$$

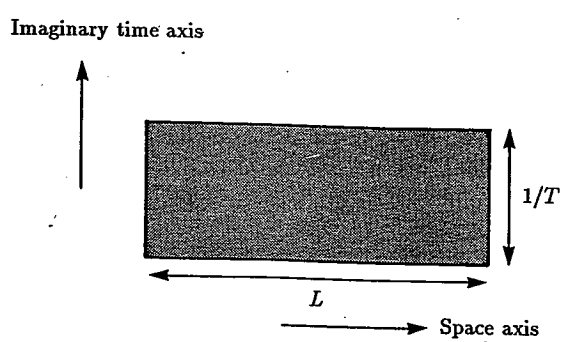


Figure 5.13 Euclidean space-time for a system of length L at temperature T. With periodic boundary conditions in imaginary time, Euclidean space-time is a cylinder.

we find the temperature dependence of u to be

$$\frac{1}{u(T)} = \frac{1}{u_0} + \frac{1}{2\pi} \ln\left(\frac{a_0 T}{v_s}\right). \tag{5.8.4}$$

Equivalently, we can write

$$u(T) = \frac{u_0}{1 + \frac{u_0}{2\pi} \ln(\frac{a_0 T}{v_s})}. \tag{5.8.5}$$

Thus, at high temperatures, $T \gg v_s/a_0$, we find that $u(T)$ becomes small

$$u(T) \approx \frac{2\pi}{\ln(\frac{a_0 T}{v_s})} \to 0 \quad \text{for} \quad T \to \infty. \tag{5.8.6}$$

In other words, the effective coupling at short distances or at high temperatures is *small*. This result is known as asymptotic freedom and, in this context, was first discussed by Polyakov [Polyakov 75].

Conversely, as the temperature is lowered, the effective coupling u becomes large (Fig. 5.14). Equation (5.8.5) exhibits an apparent divergence at the temperature T_0 where

$$T_0 \approx \frac{v_s}{a_0} e^{-\frac{2\pi}{u_0}} = \frac{v_s}{a_0} e^{-\pi s}. \tag{5.8.7}$$

The meaning of T_0 is that of the temperature at which the weak coupling (i.e. $\frac{1}{s}$) expansion breaks down. To continue down to lower temperatures, we must take into account the fact that for $T \leq T_0$, the sigma model has a large effective coupling. At this point we notice that, at large values of the coupling constant, the sigma model is disordered no matter the dimensionality of space-time. Thus we expect a finite, and short, correlation length ξ. The effective coupling should saturate due to lattice effects and the constraint $\vec{m}^2 = 1$. These ideas have been confirmed by Monte Carlo RG studies [Shenker 80].

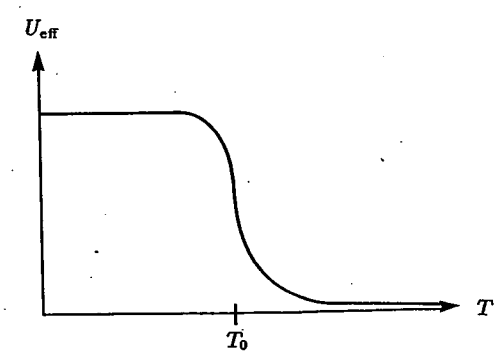


Figure 5.14 Crossover of the effective coupling $u_{eff}(T)$.

We can also use the RG to estimate the dependence of the correlation length ξ on the bare coupling constant $u_0 = \frac{2}{s}$. Under the RG, the correlation length, like all other physical observables, remains invariant. From dimensional analysis we expect ξ, which is a length scale, to have the form

$$\xi(u) = a_0 f(u) \tag{5.8.8}$$

where f is a function of u, the coupling constant *at the scale* a_0. Being an RG invariant, it must obey

$$a_0 \frac{d\xi}{da_0} = 0 \tag{5.8.9}$$

which implies that $f(u)$ satisfies the differential equation

$$\beta(u)\frac{df}{du} + f(u) = 0. \tag{5.8.10}$$

The solution to Eq. (5.8.10) is

$$f(u) = f(u')e^{-\int_{u'}^{u} \frac{dz}{\beta(z)}} = f(u')e^{2\pi(\frac{1}{u}-\frac{1}{u'})}. \tag{5.8.11}$$

where u and u' are connected by the RG flow.

Consider now the correlation length ξ at *two* different values of u, u_1 and u_2 for the *same* value of the lattice constant a_0. Let u^* be a large reference value of the coupling u. From Eqs. (5.8.8) and (5.8.11), we find

$$\xi(u_i) = a_0 f(u_i) = a_0 f(u^*)e^{-\int_{u^*}^{u_1} \frac{du}{\beta(u)}} \tag{5.8.12}$$

for both $i = 1, 2$. Thus the *ratio* of values of ξ for two different couplings and equal lattice spacing a_0 is given by

$$\frac{\xi(u_1)}{\xi(u_2)} = \frac{a_0 f(u^*)e^{-\int_{u^*}^{u_1} \frac{dz}{\beta(z)}}}{a_0 f(u^*)e^{-\int_{u^*}^{u_2} \frac{dz}{\beta(z)}}}. \tag{5.8.13}$$

Thus, we get

$$\frac{\xi(u_1)}{\xi(u_2)} = e^{-\int_{u_2}^{u_1} \frac{dz}{\beta(z)}}. \tag{5.8.14}$$

The integral can be easily evaluated to find

$$\frac{\xi(u_1)}{\xi(u_2)} = e^{(\frac{2\pi}{u_1}-\frac{2\pi}{u_2})}. \tag{5.8.15}$$

We find for the case in which $u_1 = u_0 = \frac{2}{s}$ and u_2 is large that

$$\xi(u_0) \approx \xi(u_2)e^{\pi s}. \tag{5.8.16}$$

What value should we assign to $\lim_{u_2\to\infty} \xi(u_2)$? The answer depends on whether the spin is integer or half-integer.

(a) Integer spin: In this case we do not get a topological term. As was emphasized above, the sigma model is always disordered at strong coupling. Thus, we expect $\xi(u_2) \approx a_0$ and we find a *finite* correlation length

$$\xi_0 = \xi(u_0) \approx a_0 e^{\pi s}. \tag{5.8.17}$$

There is no long-range order (i.e. no Néel state). The spectrum has a gap

$$\Delta = \frac{v_s}{\xi_0} \tag{5.8.18}$$

and the ground state is unique. Equation (5.8.17) shows that the correlation length is non-perturbative in the $\frac{1}{s}$ expansion.

(b) Half-integer spin: The sigma model coupling constant u still scales to strong coupling but the topological term remains unchanged at the value $\theta = 2\pi s \pmod{2\pi}$. However, the coupling constant $g \propto u$ is related to the spin through $s = \frac{2}{g}$. Thus strong coupling is equivalent to low spin. Hence the behavior of *all* half-integral spin chains is qualitatively identical to the spin one-half case for which $u_0 \propto 4$. The spin one-half case is *gapless*, as we saw from the Bethe-Ansatz and other approaches. Thus, $\xi(\infty)$ is still infinite. All half-integral spin chains are at a critical point with infinite correlation length. At first sight, this result seems to be paradoxical. We started with smooth configurations with well defined winding numbers and a weak coupling g. As the energy scale was lowered the effective coupling of the sigma model grew *but* the topological coupling was unaffected. Thus, at low energies, the configurations become *rough* and it is unclear what is the actual meaning of the topological term in this situation. This poses no problems for the *integer* spin chains since the topological term does not contribute in this case ($\theta = 2\pi s$). In contrast, for half-integral chains, this result simply means that *all* $s > \frac{1}{2}$ systems behave *qualitatively* in the same way as the $s = \frac{1}{2}$ case.

This result, s integer is disordered, and s half-integer is critical, is known

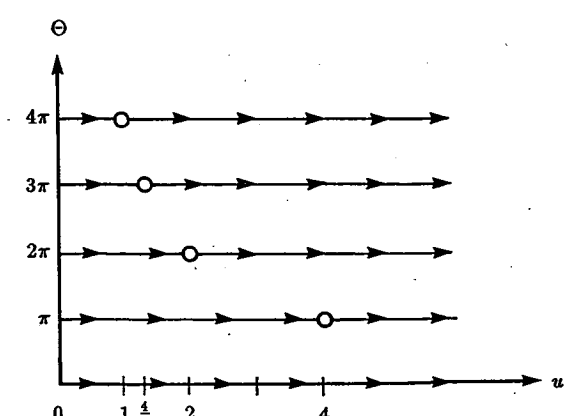


Figure 5.15 RG flows of QHA chains for $s = \frac{1}{2}, 1, \frac{3}{2}, 2$. The open circles represent their σ-model bare coupling constant $u \approx \frac{2}{s}$. They all iterate to $u^* = \infty$. The difference in their behaviour is a consequence of the presence of a $\theta = \pi$ term in the σ-model for the half-integer spin chains.

as Haldane's conjecture [Haldane 83]. It has also been checked by accurate numerical calculations using exact diagonalization on finite (but large) chains [Moreo 87], [Ziman 87] and by Green's function Monte Carlo simulations [Liang 90]. Affleck and Haldane [Affleck 87] have also found the same result using non-Abelian bosonization [Witten 84]. The RG flows are shown in Fig. 5.15.

5.9 Hopf Term or No Hopf Term?

The one-dimensional spin chains have a very unusual behavior: disorder (integer spin) or critical (half-integer spin) ground states, neutral fermions which are massless for the half-integer case and massive for s integer, etc. There is nothing in this picture which is remotely close to the physics that emerges from the mean-field theory of Chapter 3. It is then natural to ask whether or not this picture is peculiar to one-dimensional systems or if there is a natural generalization to higher dimensions. It is a trivial matter to generalize the one-dimensional formalism to the case of a square lattice. The lattice action is a simple generalization of Eq. (5.5.1). Let $\vec{r}$ span a square lattice of size $N \times N$: $\vec{r} = (x_1, x_2)$, where $x_1, x_2 = 1 \ldots N$. I will assume that N is even. The action is

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s \sum_{\vec{r}} \mathcal{S}_{\mathrm{wz}}[\vec{n}(\vec{r}\,)] - \int_0^T dx_0 \sum_{\langle \vec{r}, \vec{r}' \rangle} J s^2 \vec{n}(\vec{r}, x_0) \vec{n}(\vec{r}\,', x_0) seqnu$$

where $\vec{r}$ and $\vec{r}\,'$ are nearest neighboring sites on the square lattice. Since the square lattice is bipartite and we expect short-range Néel order, we will once again stagger the field configurations and find

$$\mathcal{S}_{\mathrm{M}}[\vec{n}] = s \sum_{\vec{r}} (-1)^{x_1 + x_2} \mathcal{S}_{\mathrm{wz}}[\vec{n}(\vec{r}\,)] + \int_0^T dx_0 \sum_{\langle \vec{r}, \vec{r}' \rangle} J s^2 \vec{n}(\vec{r}, x_0) \vec{n}(\vec{r}\,', x_0). \tag{5.9.1}$$

It is straightforward, but tedious, to derive the effective action for the slow varying fields. Once again, on the basis of symmetry, we expect a non-linear sigma model. The issue is whether or not there is a topological term in the effective action.

Before deriving the effective action by an explicit calculation, let us consider what topological terms are possible. In the $(1+1)$-dimensional case we saw that the configurations were classified in terms of an index, the topological charge, which labels the homotopy class of the configuration. The existence of such an index was guaranteed by the fact that the configurations fall into homotopy classes which form the group $\Pi_2(S_2)$ of smooth maps of the two-dimensional Euclidean space-time S_2 into the S_2 of the order parameter manifold. This homotopy group $\Pi_2(S_2)$ is isomorphic to $\mathcal{Z}$, the group of integers, i.e. the winding numbers of the topological classes. In $(2+1)$–dimensions the situation is rather different. Once again, the Euclidean space-time can be

regarded as a sphere S_3 and the configurations are maps of S_3 into S_2. However, there are no smooth solutions of the classical Euclidean equations of motion with non-trivial winding numbers. There are *singular* solutions which are known as hedgehogs (Fig. 5.16) but which have linearly divergent action. Haldane has recently speculated that these hedgehogs may become relevant if the sigma model becomes disordered by some mechanism [Haldane 88].

On the other hand, there are non-trivial configurations in Minkowski space-time (i.e. real time). Consider at some time $t = t_0$ a configuration of sigma-model fields identical to one of the instantons of Section 5.6. Now it represents the snapshot of an eigenstate and it is called a soliton. Thus the *configuration space* of a two-dimensional quantum non-linear sigma model is also a sphere S_2 and is usually denoted by $\Omega_2 S_2$. Consider now the real-time evolution of such a state with periodic boundary conditions in time, i.e. we consider histories in which the initial state is the same as the final state. Thus, a history is a closed curve in the configuration space $\Omega_2 S_2$. In quantum mechanics we are told to sum over all histories and to assign a phase to each history, i.e. to each curve in $\Omega_2 S_2$. Since a phase is an element of S_1 (the unit circle) we have constucted the set of maps $\Pi_1(\Omega_2 S_2)$. However, we know that the configurations at any given time are maps of S_2 (space) into S_2 (field), i.e. Homotopy classes of $\Pi_2(S_2)$ which we saw was isomorphic to the group of integers $\mathcal{Z}$. Hence the configuration space $\Omega_2 S_2$ is decomposed into a disjoint union of path-connected pieces, each characterized by the winding number or soliton number Q. Thus each disconnected piece of the Hilbert space will have a separate time evolution and will have to be summed with separate phases. Since the classical paths are continuous curves in $\Omega_2 S_2$ classified by $\Pi_3(S_2) = \mathcal{Z}$, the relevant issue is now what topolgical invariant is associated with such histories.

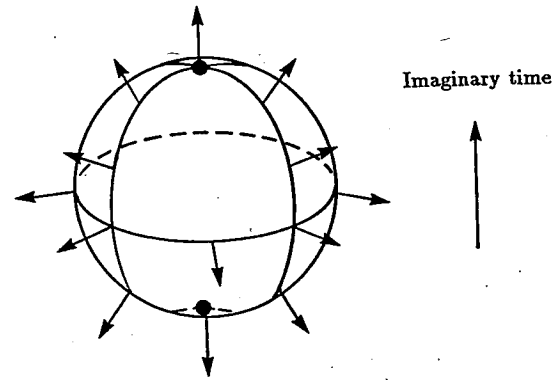


Figure 5.16 A hedgehog.

Consider a history of the order parameter field $\vec{m}(\vec{x},t)$ in $(2+1)$ dimensions. We can define a *topological current* J_μ by

$$J_\mu = \frac{1}{8\pi}\epsilon_{\mu\nu\lambda}\epsilon_{abc}m_a\partial^\nu m_b\partial^\lambda m_c \tag{5.9.2}$$

with $\mu = 0,1,2$ and $a,b,c,= 1,2,3$. The topological current J_μ is clearly conserved

$$\partial^\mu J_\mu = 0. \tag{5.9.3}$$

Therefore the total *topological charge* $Q = \int d^2x J^0(\vec{x},t)$ is constant in time

$$Q = \int d^2x J^0(\vec{x},t) = \int d^2x \frac{1}{8\pi}\epsilon_{0ij}\epsilon_{abc}m_a\partial^i m_b\partial^j m_c. \tag{5.9.4}$$

Clearly Q is identical to the winding number $\mathcal{Q}$ of Eq. (5.6.4).

Consider now a soliton state with $\mathcal{Q} = 1$ (Fig. 5.8). Imagine a time evolution in which the soliton rotates slowly around its center and executes exactly n turns during its lifespan. Each point on the equator of the soliton traces a curve ("worldline") which wraps n-times around the other curves traced by the other points (Fig. 5.17). An easy way to compute the winding number of this history is to imagine that each worldine is a wire carrying the unit of current. As the soliton rotates, the worldlines ("wires") are *braided*. The natural topological invariant is the *linking number* of these worldlines (Fig. 5.18). If we denote by $\vec{j}$ the current carried by the wires and by $\vec{B}$ the magnetostatic field they create, the linking number is simply given by Ampere's law

$$\int d^3x \vec{j}\cdot\vec{B} = 2\pi n \tag{5.9.5}$$

where n is the number of turns.

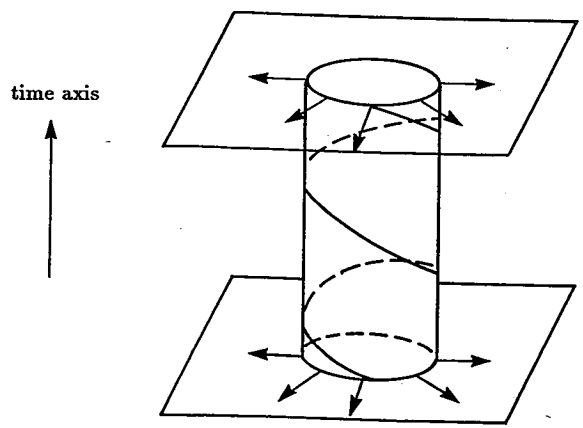


Figure 5.17 A world tube with linking number +2.

We can make this analogy more precise by using the *Hopf map* which maps the three sphere S_3 onto S_2. Let z_1 and z_2 be two complex number satisfying

$$|z_1|^2 + |z_2|^2 = 1. \tag{5.9.6}$$

Clearly (z_1, z_2) span S_3. Define now the *spinor* $z_\alpha, (\alpha = 1,2)$. The order parameter field $\vec{m}$ is related to z_α through the map

$$m^a = z^*_\alpha \sigma^a_{\alpha\beta} z_\beta \tag{5.9.7}$$

where $\{\sigma^a\}_{a=1,2,3}$ are the Pauli matrices. The order parameter $\vec{m}$ also satisfies $\vec{m}^2 = 1$. This is the Hopf map.

It is clear that (z_1, z_2) has three independent parameters whereas $\vec{m}$ only has two. But one of these parameters, or degrees of freedom, is unobservable since a global chage of phase of the spinor z_α

$$\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \to e^{i\phi}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \tag{5.9.8}$$

does not lead to any observable effects since $\vec{m}$ is *invariant* under such *gauge transformations*.

Furthermore, the action of the non-linear sigma model itself can be written in terms of the spinor field z_α. This is the CP^1 model. To simplify matters I will consider the problem with spin-wave velocity $v_s = 1$. Let z_α be a CP^1

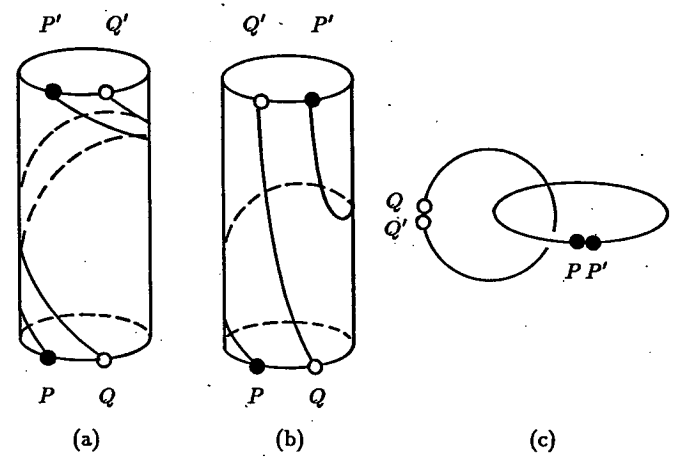


Figure 5.18 (a) Two world lines PP' and QQ' with linking number +1. (b) Smooth deformation of the world line PP' and QQ'. (c) Periodic boundary conditions in time are enforced, P and P' as well as Q and Q' are identified and the world lines form a braid.

field and A_μ an unconstrained gauge field, with Lagrangian density

$$\mathcal{L}_{\mathrm{CP}^1} = \frac{1}{2g}|D_\mu z|^2 \tag{5.9.9}$$

where g is a coupling constant and D_μ is the covariant derivative

$$D_\mu = \partial_\mu - iA_\mu. \tag{5.9.10}$$

The functional integral is

$$Z = \int \mathcal{D}\bar{z}\mathcal{D}z\delta(|z|^2 - 1)\int \mathcal{D}A e^{iS_{\mathrm{CP}^1}[z,A]}. \tag{5.9.11}$$

Since $\mathcal{L}_{\mathrm{CP}^1}$ is quadratic in A_μ, it can be integrated out exactly by a saddle point calculation. The saddle point condition

$$\frac{\delta\mathcal{L}_{\mathrm{CP}^1}}{\delta A_\mu} = 0 \tag{5.9.12}$$

determines the gauge field in terms of the CP^1 field

$$A_\mu = \frac{i}{2}(z^*_\alpha\partial_\mu z_\alpha - z_\alpha\partial_\mu z^*_\alpha) \equiv -\frac{i}{2}z^*_\alpha\overleftrightarrow{\partial_\mu} z_\alpha. \tag{5.9.13}$$

By substituting Eq. (5.9.14) in the Lagrangian density, Eq. (5.9.10), one finds

$$\frac{1}{2g}(\partial_\mu\vec{m})^2 = \frac{1}{g}|D_\mu z|^2. \tag{5.9.14}$$

In other words, the CP^1 model and the $O(3)$ non-linear sigma model are equivalent.

The topological invariant, or *Hopf invariant*, has a very simple and natural form in terms of the vector potentials A_μ. Consider a term in the Lagrangian density of the form

$$\mathcal{L}_{\mathrm{CS}} = \frac{\theta}{4\pi}\epsilon_{\mu\nu\lambda}A^\mu F^{\nu\lambda} \tag{5.9.15}$$

which is known as a Chern-Simons term. The gauge field A_μ is *constrained* to be given by Eq. (5.9.14) and its field strength can be related back to the sigma model field $\vec{m}$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \equiv \vec{m}\cdot(\partial_\mu\vec{m}\times\partial_\nu\vec{m}). \tag{5.9.16}$$

Thus, the flux associated with the gauge field A_μ is simply related to the topological current. The *Hopf invariant* H is simply

$$H = \frac{\theta}{8\pi^2}\int d^3x\epsilon_{\mu\nu\lambda}A^\mu F^{\nu\lambda} \tag{5.9.17}$$

with A_μ and $F_{\nu\lambda}$ given by Eqs. (5.9.14) and (5.9.17). We will see in chapter 7 that a non-zero value of θ will change the statistics of the solitons.

But, is there a Hopf term in the effective action of the quantum Heisenberg antiferromagnet in two dimensions? The only way to determine that is to compute the effective action carefully. [Dzyaloshinskii 88] and [Wiegmann 88] have conjectured that the effective action has a Hopf term with $\theta = 2\pi s$. This is a subtle business since Wu and Zee have shown that, in its CP^1 form, the

Hopf term is a total derivative which does not alter the equations of motion but changes the spin and statistics of the topological excitations [Wu 84].

Let us first derive the effective action, following the methods of Fradkin and Stone [Fradkin 88]. The result will be a sigma model *without* a topological term [Fradkin 88], [Haldane 88], [Wen 88], [Ioffe 88], [Dombre 88].

First, we need to integrate out the fast degrees of freedom. We write

$$\vec{n}(\vec{r}\,) = m(\vec{r}\,) + (-1)^{x_1+x_2}a_0\vec{l}(\vec{r}\,). \tag{5.9.18}$$

Following the *same* procedure which in the one-dimensional case led to a sigma model *with* a topological term (see Eq. (5.5.11)), we find

$$\mathcal{L}^M_{\mathrm{eff}}(\vec{m},\vec{l}) = -\frac{Js^2}{2}\left((\partial_i\vec{m})^2 + 8\vec{l}^{\,2}\right) + \frac{s}{a_0}\vec{l}\cdot(\vec{m}\times\partial_0\vec{m}). \tag{5.9.19}$$

If we now proceed to integrate out the fast modes, the $\vec{l}$ field, we find a non-linear sigma model *without* a topological term. The (bare) coupling constant and (bare) spin-wave velocity are given by (see Eqs. (5.5.14) and (5.5.15))

$$\begin{aligned} g &= \sqrt{2}a_0\frac{2}{s} \\ v_s &= \sqrt{2}2a_0Js. \end{aligned} \tag{5.9.20}$$

The terms, which in the one-dimensional case gave rise to the toplogical term, now have cancelled each other out. The reason for this cancellation can be traced back to the staggered character of the Néel state. Naively, we expect that each row will make a contribution similar to the one-dimensional result. But neighboring rows are staggered in the opposite way. The result is that the terms originating from each neighboring rows, now, effectively cancel out. We are assuming a lattice with even rows and columns. In the case of an *odd* number of rows, we may get a non zero contribution from the last row. However, this is a boundary condition effect which, incidentally, was not needed in the case of the chains. But we do expect to see changes in the spectrum of elementary excitations if we change the boundary conditions.

The argument which led to the cancellation is a bit too naive and maybe dangerous. We know from the work of Wu and Zee that, at least in the CP^1 representation, the Hopf term is a total derivative. Thus, a local cancellation is not a sufficient argument for the study of a *global* effect. Slowly varying configurations may have an accumulated effect near the boundaries and yield a non-zero answer. We can dispel these fears by computing the alternating sum Φ

$$\Phi = s\sum_{\vec{r}}(-1)^{x_1+x_2}\mathcal{S}_{\mathrm{wz}}[\vec{n}(\vec{r}\,)] \tag{5.9.21}$$

for a configuration which, in the continuum limit, has soliton number $Q = 1$. If we let this soliton configuration rotate slowly around its center such that it turns exactly once during its history, the history of this configuration should have Hopf number or linking number +1. We should choose a lattice configuration which, in the limit of soliton radius r_s *large* compared with the lattice spacing a_0 should go smoothly over to the continuum soliton. Any

soliton profile should do the job. For instance, we can imagine a configuration obtained by a stereographic configuration (Fig. 5.19). The area swept by each spin is $a = 2\pi R^2(1-\cos\theta)$. Thus the sum $\Phi(N)$, for a system of size $2N \times 2N$, is given by

$$\Phi(N) = \sum_{n,m=-N+1}^{N} 2\pi s R^2 \left(1 - \cos\theta(n,m)\right)(-1)^{n+m}. \tag{5.9.22}$$

The sphere has radius R and its south pole, which has coordinates (α_1, α_2), is in the first unit cell. The radius of the soliton r_s is equal to the diameter $2R$ of the sphere, if we define the radius as the location in which the spins are orthogonal to the asymptotic configuration at spatial infinity. Hence we find

$$\Phi(N) = \sum_{n,m=-N+1}^{N} \frac{16\pi s R^4 e^{i\pi(n+m)}}{4R^2 + (n-\alpha_1)^2 + (m-\alpha_2)^2}. \tag{5.9.23}$$

In the thermodynamic limit, $N \to \infty$, and by making use of the Poisson Summation Formula

$$\sum_{n=-\infty}^{+\infty} f(n) = \sum_{n=-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{dk}{2\pi} e^{i2\pi kn} f(k), \tag{5.9.24}$$

we get for $\Phi \equiv \lim_{N\to\infty} \Phi(N)$

$$\Phi = \int \frac{d^2k}{(2\pi)^2} \sum_{\vec{n}} \frac{16\pi s R^4 e^{i2\pi(\vec{k}+\vec{\alpha})\cdot(\vec{n}+\vec{G})}}{4R^2 + \vec{k}^2} \tag{5.9.25}$$

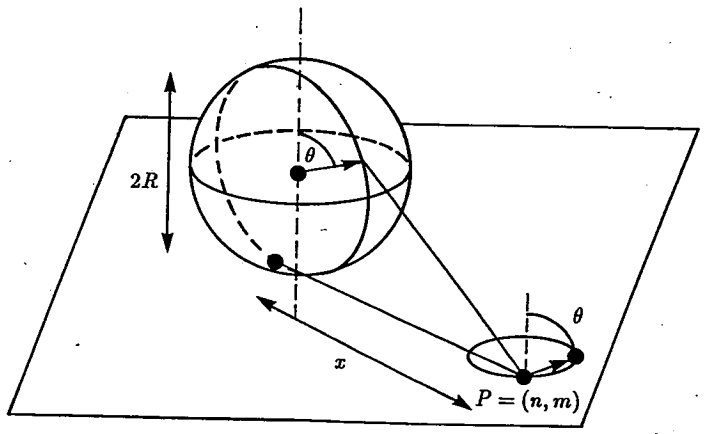


Figure 5.19 A soliton configuration can be generated using a stereographic projection. The spin is paralell-transported from the sphere to point $P = (n, m)$. Its history is pictured as an ellipse.

where $\vec{G} = (\frac{1}{2}, \frac{1}{2})$. In the limit $R \to \infty$ it is easy to see that Φ is exponentially small since we can write

$$\Phi = \sum_{n_1,n_2=-\infty}^{+\infty} 8sR^4 e^{i2\pi\vec{\alpha}\cdot(\vec{n}+\vec{G})} K_0(\pi r_s|\vec{n}+\vec{G}|) \tag{5.9.26}$$

where $K_0(x)$ is a modified Bessel function. Thus, for $r_s \gg 1$, we may keep just the leading terms:

$$\Phi \approx 4s \left(\frac{r_s}{\sqrt{2}}\right)^{5/2} e^{-\frac{2\pi}{\sqrt{2}}r_s + i2\pi(\alpha_1+\alpha_2)} \cos(\pi\alpha_1)\cos(\pi\alpha_2). \tag{5.9.27}$$

This expression vanishes exponentially fast for solitons with radii $r_s \gg \frac{1}{\pi\sqrt{2}}$. Notice that even fairly small solitons with radii $r_s \approx 1$ are large according to this criterion. We must conclude that if we expect to see Néel order (even if this was true only at short distances!) the effective theory at long distances is given by a non-linear sigma model with renormalized coupling constant and spin-wave velocity. Phenomenologically this is what the experiments in $La_2Cu_1O_4$ seem to indicate [Chakravarty 88]. In Section 5.7 we calculated the one-loop β-function for the non-linear sigma model in $(2+\epsilon)$ dimensions (here 2 means $(1+1)$). We found the result (see Eq. (5.7.21))

$$\beta(u) = -\epsilon u + \frac{u^2}{2\pi}. \tag{5.9.28}$$

For space-time dimensions $d > 2$, the fixed point at the origin is *stable*. This means that if the bare dimensionless coupling constant u is sufficiently small, the effective coupling flows toward the $u = 0$ fixed point and we have a Néel state with weakly coupled spin-waves. Equation (5.7.21) has another fixed point at $u^* \approx 2\pi\epsilon$, which is *infrared unstable*. This fixed point is the location of a second-order phase transition (in terms of the coupling constant). Beyond this fixed point, i.e. for $u > u^*$, the effective coupling flows toward the $u = \infty$ fixed point just as in the (1+1)-dimensional case. However, we do not have a topological term anymore. Thus, we must conclude that, for $u > u^*$ the system is disordered at distances longer than some correlation length $\xi \sim |u-u^*|^{-\nu}$ $\left(\nu = \frac{1}{d-2} + \mathcal{O}(d-2)\right)$ and Néel-like order at scales between the lattice constant and ξ. Such a state is a *zero-temperature quantum paramagnet* (QP), i.e. a paramagnetic state driven purely by quantum fluctuations and in the absence of thermal fluctuations. A finite correlation length without long-range order means that the ground state is unique and there is a gap $\Delta = v_s/\xi$ for the elementary excitations ("spin-waves").

The theory described here, which is based in the $(2+\epsilon)$ expansion, is too crude to reliably predict the value of u^*. Since we saw that our approximations were equivalent to (a resummation of) the $\frac{1}{s}$ expansion, we must also conclude that u^* cannot be calculated with confidence from the $\frac{1}{s}$ expansion either. Qualitatively, we should still expect a non-trivial fixed point for $\epsilon = 1$. The perturbative β-function predicts that for $\epsilon \approx 1$ even $s = \frac{1}{2}$ on a square lattice is on the Néel side of the phase transition although not far from it. This

result appears to be consistent with existent experimental data on quasi-two-dimensional systems believed to be reasonably well described by the $s = \frac{1}{2}$ quantum Heisenberg antiferromagnet such as $La_2Cu_1O_4$. Experimentally [Shirane 87] one sees a Néel state but with a magnetic moment about fifty percents of its classical value. The dynamical structure factor predicted by the σ-model [Chakravarty 88] is also confirmed by these experiments. Numerical calculations on two-dimensional quantum Heisenberg models also exhibit a similar behavior [Liang 88], [Liang 90].

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